

COM 307
Machine Learning and Data Mining
Project 1 write-up

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1. What methods/scenarios did well?

Maximum Likelihood Estimation (MLE) and all Bayesian Scenarios (2, 3, and 4) performed exceptionally well on the largest dataset (Dataset C, 2000 rolls) because there was enough data. For the smaller datasets, Scenario 4 performed the best because its prior assumptions $p(\theta=0.5)=52\%$ closely matched the true underlying distribution, helping the model make an accurate prediction even when the size of data was not big enough.

2. What methods/scenarios did not?

MLE kind of struggled with the 20-roll dataset because it relies on the observed data, which is highly susceptible to random noise in small sample sizes (MLE likely predicted around 55% instead of 50%). Scenario 2 also struggled on the 20-roll dataset, failing to definitively choose between $\theta = 0.5$ and $\theta = 0.6$ (giving them both around 50% probability) because prior knowledge it had was not enough.

3. Did the methods/scenarios all perform well for some datasets?

Yes, for Dataset C (2000 rolls).

As we discussed in class, when the amount of observed data approaches infinity, the data completely overpowers the prior assumptions. Probability of 6 was 50.5% in scenario 1, which is pretty close to our 50% setting for the `die_throw()` function. Scenarios 2, 3, and 4 all converged to a 100% probability for $\theta = 0.5$, regardless of what prior they started with. We can also see a clear general trend that the probability for $\theta = 0.5$ increases as the size of prior dataset increases. (50.1% to 56.3% to 65.1%); (83.9% to 87.0% to 90.6%) and then (100%, 100%, 100%)

4. Were there other datasets where it made a bigger difference what method/scenario was used?

Yes, the scenario made a massive difference for Dataset A (20 rolls). When the sample size is small, the model leans heavily on its prior assumptions. Scenario 2 was completely unsure of the true probability, while Scenario 4 was able to better predict the correct θ (65% probability) entirely because of its strong, accurate prior.

I used an AI assistant (google search gemini) to understand and implement the python Decimal module to prevent floating point underflow errors on the 2000 dataset.

Table below includes the summary of the 4 scenarios

Dataset (Actual Data)	Scenario 1: MLE (Probability of 6)	Scenario 2: Bayesian	Scenario 3: Bayesian	Scenario 4: Bayesian
Dataset A (20 Rolls) 11 sixes, 9 non-sixes	55.0%	Theta 1/6: 0.0% Theta 0.5: 50.1% Theta 0.6: 49.9%	Theta 1/6: 0.0% Theta 0.5: 56.3% Theta 0.6: 43.7%	Theta 1/6: 0.0% Theta 0.5: 65.1% Theta 0.6: 34.9%
Dataset B (200 Rolls) 106 sixes, 94 non-sixes	53.0%	Theta 1/6: 0.0% Theta 0.5: 83.9% Theta 0.6: 16.1%	Theta 1/6: 0.0% Theta 0.5: 87.0% Theta 0.6: 13.0%	Theta 1/6: 0.0% Theta 0.5: 90.6% Theta 0.6: 9.4%
Dataset C (2000 Rolls) 1010 sixes, 990 non-sixes	50.5%	Theta 1/6: 0.0% Theta 0.5: 100.0% Theta 0.6: 0.0%	Theta 1/6: 0.0% Theta 0.5: 100.0% Theta 0.6: 0.0%	Theta 1/6: 0.0% Theta 0.5: 100.0% Theta 0.6: 0.0%